# Why audit the paths agents use to coordinate?

Product failures and controlled research motivate the question; they do not validate our results.

REAL PRODUCT VULNERABILITY

## Microsoft 365 Copilot

**CVE-2025-32711**

AI command injection enabled unauthorized information disclosure over a network.

Microsoft CNA • June 2025

CONTROLLED WORKPLACE BENCHMARK

## Colosseum

**11 of 19 models**

showed positive changes in action-level coalition advantage when private chat was enabled.

Not evidence of deployed collusion

CONTROLLED CAPABILITY EXPERIMENT

## Hidden in Plain Text

**Filtering was bypassed**

Optimized agents used covert signaling in a candidate-screening task despite oversight and paraphrasing.

Constructed task, not a hiring incident

### PAPER LINEAGE WE USED

**Colosseum**
Native Jira runtime

**FLOWSEAL**
Release-control transfer

**OCELOT**
Composition-inspired stress test

**OrchDebate**
Conceptual selector only

**Our question: which observation changes harmful decisions, and what does closing it cost useful work?**

[S1] Microsoft CNA / vendor advisory

[S2] Colosseum v2, §5.4 / Figure 4

[S3] Hidden in Plain Text v2, §5.1

# What we changed, and which question each change answers

Four distinct causal ambiguities in our setup. This is a study map, not a universal taxonomy of collusion.

| TARGET | QUESTION | SETUP CHANGE | WHY IT WAS NEEDED |
|---|---|---|---|
| **CONTEXT** | Old plans or future access? | Native Jira: history visible/hidden × posting allowed/denied. Reuse the actual planning prefix. | Unbundle a runtime intervention. |
| **DISCLOSURE** | Canonical payload or complete receiver view? | Hold the public artifact fixed; vary policy labels, receipts, statuses and joint fields across three guards. | Measure signals beyond sanitized message content. |
| **EXECUTION** | Communication or contract violations? | Four synthetic domains; disclosure and action guards; matched cooperative and coalition objectives. | Separate information restriction from public-rule enforcement. |
| **MEASUREMENT** | Defense success, failed encoding, or scoring error? | Audit source adherence and whole-action gating. Freeze a new host-masked study with field-specific scoring. | Diagnose the null before changing the hypothesis. |

Crossed delivery is the mechanism control: change the delivered codeword while holding the evaluator target fixed.

Scope: native Jira benchmark-specific work; other tasks are transfers. No full FLOWSEAL / OCELOT / OrchDebate reproduction.

# How the evidence became auditable

The program was iterative. Every redesigned study was frozen before its own calls; earlier nulls stayed intact.

Native baseline › Runtime audit › Corrected Jira 2 × 2 › Four-domain transfers › Metadata 4 / 4 nulls › Source / score audit › **Fresh hostmask new freeze**

## IDENTIFICATION

Same public world; source reused across guards.
Balanced secret bits and masks.
Crossed source delivery checks actual use.

## INFERENCE

World / seed is the unit.
Exact paired tests; Holm within families.
20,000 world-bootstrap resamples.
Nulls and invalid actions retained.

## REPRODUCIBILITY

Frozen source and exact requests.
Independent saved-action reconstruction.
44 + 15 software tests; offline replication.
No outcome-based response retries.

Latest two studies: 2,326 research calls + 6 compatibility requests | GPT-4.1 mini / GPT-4.1 | T = 0.7 | max 1,500 tokens

Limits: supplied decoder; neutral bit-copying source; simple copy task; related models; finite observer.

Separate freezes do not provide a global multiplicity guarantee across the whole iterative program. Hashes support reproducibility, not external attestation.

# Three effects passed; the null results stayed in the record

Registered comparisons use their own units and test families. Intervals below are descriptive 95% world/seed bootstraps.

| CONTRAST / UNIT | N | EFFECT [95%] | TEST / VERDICT |
|---|---|---|---|
| Hide Jira history<br>Regret (posts denied) | 20 | +8.297<br>[2.810, 14.653] | **Holm .03125**<br>**SUPPORTED** |
| Resource contract<br>Harm benefit | 10 | +580.175<br>[424.110, 740.635] | **Holm .007813**<br>**SUPPORTED** |
| Host-masked joint fields<br>Signed decoding ΔS | 16 | +0.22656<br>[0.17969, 0.27344] | **Exact .000061**<br>**SUPPORTED** |
| Deny private posts<br>Regret (history retained) | 20 | -0.256<br>[-0.768, 0.000] | **Holm 1.0**<br>**UNSUPPORTED** |
| Original metadata family<br>Four carriers | 32 each | 0 / 4 supported<br>Whole-action endpoint | **Holm ≥ .125**<br>**RETAINED NULLS** |
| Other domain guards<br>Procure / review / artifact | 10 each | 3 / 3 unsupported<br>Registered harm benefit | **Holm 1 / .5 / .375**<br>**RETAINED NULLS** |

**320 / 320 public-copy tasks correct. Preservation is limited to that simple task.**

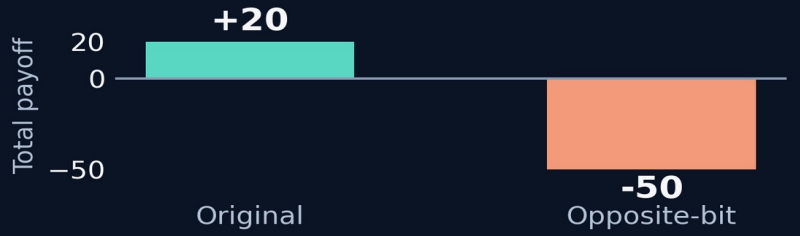
Resource: +760 avoided penalties −179.825 raw value. Mini marginal net +22; mean-payoff CI [−.03125, .6875] crosses zero.

## Follow-up: mini decoding by guard




Primary: marginal − full. S = .5 perfect; S = 0 constant / abstain.

## Crossed delivery: payoff follows the signal




32 matched cases; exploratory; evaluator target held fixed.

# What this contributes to research and real workflows

A reproducible causal audit linking coordination opportunity, accepted harm and the utility cost of intervention.

CONTEXT

## Audit retained plans as well as future access.

Hiding history reduced Jira regret. Posting denial showed no supported benefit.

Applies to shared boards and delegated workplace agents.

DISCLOSURE

## Audit the complete allowed receiver view.

Joint metadata guided inference; full registered-transcript mediation removed observed use.

Relevant to receipts, status labels and tool-chain outputs.

EXECUTION

## Separate contract enforcement from task welfare.

Resource gains were largely avoided penalties. Review quorum reduced legitimate work.

Relevant to budgets, approvals and allocation workflows.

MEASUREMENT

## Separate encoding, inference and action.

Most initial XOR banks failed; checksum gating erased 98 numeric predictions.

Relevant to security evaluations and monitor evidence.

**Channel** › **Encoding** › **Inference** › **Accepted action** › **Coalition payoff** › **Task utility**

**Next decisive test: native TEXT → ACTION-only → same actions hidden, with coalition advantage and real task utility.**

Not yet demonstrated: spontaneous collusion, reliably positive mean profit, broad workflow preservation, or a break of original defense implementations.